



12080CH07

INTEGRALS

❖ *Just as a mountaineer climbs a mountain – because it is there, so a good mathematics student studies new material because it is there. — JAMES B. BRISTOL* ❖

7.1 Introduction

Differential Calculus is centred on the concept of the derivative. The original motivation for the derivative was the problem of defining tangent lines to the graphs of functions and calculating the slope of such lines. Integral Calculus is motivated by the problem of defining and calculating the area of the region bounded by the graph of the functions.

If a function f is differentiable in an interval I , i.e., its derivative f' exists at each point of I , then a natural question arises that given f' at each point of I , can we determine the function? The functions that could possibly have given function as a derivative are called anti derivatives (or primitive) of the function. Further, the formula that gives all these anti derivatives is called the *indefinite integral* of the function and such process of finding anti derivatives is called integration. Such type of problems arise in many practical situations. For instance, if we know the instantaneous velocity of an object at any instant, then there arises a natural question, i.e., can we determine the position of the object at any instant? There are several such practical and theoretical situations where the process of integration is involved. The development of integral calculus arises out of the efforts of solving the problems of the following types:

- (a) the problem of finding a function whenever its derivative is given,
- (b) the problem of finding the area bounded by the graph of a function under certain conditions.

These two problems lead to the two forms of the integrals, e.g., indefinite and definite integrals, which together constitute the *Integral Calculus*.



G .W. Leibnitz
(1646 -1716)

There is a connection, known as the **Fundamental Theorem of Calculus**, between indefinite integral and definite integral which makes the definite integral as a practical tool for science and engineering. The definite integral is also used to solve many interesting problems from various disciplines like economics, finance and probability.

In this Chapter, we shall confine ourselves to the study of indefinite and definite integrals and their elementary properties including some techniques of integration.

7.2 Integration as an Inverse Process of Differentiation

Integration is the inverse process of differentiation. Instead of differentiating a function, we are given the derivative of a function and asked to find its primitive, i.e., the original function. Such a process is called *integration* or *anti differentiation*.

Let us consider the following examples:

We know that $\frac{d}{dx}(\sin x) = \cos x$... (1)

$$\frac{d}{dx}\left(\frac{x^3}{3}\right) = x^2 \quad \dots (2)$$

and $\frac{d}{dx}(e^x) = e^x$... (3)

We observe that in (1), the function $\cos x$ is the derived function of $\sin x$. We say that $\sin x$ is an anti derivative (or an integral) of $\cos x$. Similarly, in (2) and (3), $\frac{x^3}{3}$ and e^x are the anti derivatives (or integrals) of x^2 and e^x , respectively. Again, we note that for any real number C , treated as constant function, its derivative is zero and hence, we can write (1), (2) and (3) as follows :

$$\frac{d}{dx}(\sin x + C) = \cos x, \quad \frac{d}{dx}\left(\frac{x^3}{3} + C\right) = x^2 \quad \text{and} \quad \frac{d}{dx}(e^x + C) = e^x$$

Thus, anti derivatives (or integrals) of the above cited functions are not unique. Actually, there exist infinitely many anti derivatives of each of these functions which can be obtained by choosing C arbitrarily from the set of real numbers. For this reason C is customarily referred to as **arbitrary constant**. In fact, C is the **parameter** by varying which one gets different anti derivatives (or integrals) of the given function.

More generally, if there is a function F such that $\frac{d}{dx}F(x) = f(x)$, $\forall x \in I$ (interval), then for any arbitrary real number C , (also called *constant of integration*)

$$\frac{d}{dx}[F(x) + C] = f(x), \quad x \in I$$

Thus, $\{F + C, C \in \mathbf{R}\}$ denotes a family of anti derivatives of f .

Remark Functions with same derivatives differ by a constant. To show this, let g and h be two functions having the same derivatives on an interval I .

Consider the function $f = g - h$ defined by $f(x) = g(x) - h(x)$, $\forall x \in I$

Then $\frac{df}{dx} = f' = g' - h'$ giving $f'(x) = g'(x) - h'(x) \quad \forall x \in I$

or $f'(x) = 0$, $\forall x \in I$ by hypothesis,

i.e., the rate of change of f with respect to x is zero on I and hence f is constant.

In view of the above remark, it is justified to infer that the family $\{F + C, C \in \mathbf{R}\}$ provides all possible anti derivatives of f .

We introduce a new symbol, namely, $\int f(x) dx$ which will represent the entire class of anti derivatives read as the indefinite integral of f with respect to x .

Symbolically, we write $\int f(x) dx = F(x) + C$.

Notation Given that $\frac{dy}{dx} = f(x)$, we write $y = \int f(x) dx$.

For the sake of convenience, we mention below the following symbols/terms/phrases with their meanings as given in the Table (7.1).

Table 7.1

Symbols/Terms/Phrases	Meaning
$\int f(x) dx$	Integral of f with respect to x
$f(x)$ in $\int f(x) dx$	Integrand
x in $\int f(x) dx$	Variable of integration
Integrate	Find the integral
An integral of f	A function F such that $F'(x) = f(x)$
Integration	The process of finding the integral
Constant of Integration	Any real number C , considered as constant function

We already know the formulae for the derivatives of many important functions. From these formulae, we can write down immediately the corresponding formulae (referred to as standard formulae) for the integrals of these functions, as listed below which will be used to find integrals of other functions.

Derivatives

$$(i) \quad \frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n ;$$

Particularly, we note that

$$\frac{d}{dx} (x) = 1 ;$$

$$(ii) \quad \frac{d}{dx} (\sin x) = \cos x ;$$

$$(iii) \quad \frac{d}{dx} (-\cos x) = \sin x ;$$

$$(iv) \quad \frac{d}{dx} (\tan x) = \sec^2 x ;$$

$$(v) \quad \frac{d}{dx} (-\cot x) = \operatorname{cosec}^2 x ;$$

$$(vi) \quad \frac{d}{dx} (\sec x) = \sec x \tan x ;$$

$$(vii) \quad \frac{d}{dx} (-\operatorname{cosec} x) = \operatorname{cosec} x \cot x ;$$

$$(viii) \quad \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} ;$$

$$(ix) \quad \frac{d}{dx} (-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}} ;$$

$$(x) \quad \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2} ;$$

$$(xi) \quad \frac{d}{dx} (e^x) = e^x ;$$

Integrals (Anti derivatives)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int dx = x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

$$\int e^x dx = e^x + C$$

$$(xii) \quad \frac{d}{dx}(\log |x|) = \frac{1}{x};$$

$$\int \frac{1}{x} dx = \log |x| + C$$

$$(xiii) \quad \frac{d}{dx} \left(\frac{a^x}{\log a} \right) = a^x;$$

$$\int a^x dx = \frac{a^x}{\log a} + C$$

 **Note** In practice, we normally do not mention the interval over which the various functions are defined. However, in any specific problem one has to keep it in mind.

7.2.1 Some properties of indefinite integral

In this sub section, we shall derive some properties of indefinite integrals.

- (I) The process of differentiation and integration are inverses of each other in the sense of the following results :

$$\frac{d}{dx} \int f(x) dx = f(x)$$

and

$$\int f'(x) dx = f(x) + C, \text{ where } C \text{ is any arbitrary constant.}$$

Proof Let F be any anti derivative of f , i.e.,

$$\frac{d}{dx} F(x) = f(x)$$

Then

$$\int f(x) dx = F(x) + C$$

Therefore

$$\begin{aligned} \frac{d}{dx} \int f(x) dx &= \frac{d}{dx} (F(x) + C) \\ &= \frac{d}{dx} F(x) = f(x) \end{aligned}$$

Similarly, we note that

$$f'(x) = \frac{d}{dx} f(x)$$

and hence

$$\int f'(x) dx = f(x) + C$$

where C is arbitrary constant called constant of integration.

- (II) Two indefinite integrals with the same derivative lead to the same family of curves and so they are equivalent.

Proof Let f and g be two functions such that

$$\frac{d}{dx} \int f(x) dx = \frac{d}{dx} \int g(x) dx$$

or
$$\frac{d}{dx} \left[\int f(x) dx - \int g(x) dx \right] = 0$$

Hence $\int f(x) dx - \int g(x) dx = C$, where C is any real number (Why?)

or
$$\int f(x) dx = \int g(x) dx + C$$

So the families of curves $\left\{ \int f(x) dx + C_1, C_1 \in \mathbf{R} \right\}$

and $\left\{ \int g(x) dx + C_2, C_2 \in \mathbf{R} \right\}$ are identical.

Hence, in this sense, $\int f(x) dx$ and $\int g(x) dx$ are equivalent.

 **Note** The equivalence of the families $\left\{ \int f(x) dx + C_1, C_1 \in \mathbf{R} \right\}$ and $\left\{ \int g(x) dx + C_2, C_2 \in \mathbf{R} \right\}$ is customarily expressed by writing $\int f(x) dx = \int g(x) dx$, without mentioning the parameter.

(III)
$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

Proof By Property (I), we have

$$\frac{d}{dx} \left[\int [f(x) + g(x)] dx \right] = f(x) + g(x) \quad \dots (1)$$

On the otherhand, we find that

$$\begin{aligned} \frac{d}{dx} \left[\int f(x) dx + \int g(x) dx \right] &= \frac{d}{dx} \int f(x) dx + \frac{d}{dx} \int g(x) dx \\ &= f(x) + g(x) \end{aligned} \quad \dots (2)$$

Thus, in view of Property (II), it follows by (1) and (2) that

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx.$$

(IV) For any real number k ,
$$\int k f(x) dx = k \int f(x) dx$$

Proof By the Property (I), $\frac{d}{dx} \int k f(x) dx = k f(x)$.

$$\text{Also } \frac{d}{dx} \left[k \int f(x) dx \right] = k \frac{d}{dx} \int f(x) dx = k f(x)$$

Therefore, using the Property (II), we have $\int k f(x) dx = k \int f(x) dx$.

(V) Properties (III) and (IV) can be generalised to a finite number of functions $f_1, f_2, \dots, f_n$ and the real numbers, $k_1, k_2, \dots, k_n$ giving

$$\begin{aligned} & \int [k_1 f_1(x) + k_2 f_2(x) + \dots + k_n f_n(x)] dx \\ &= k_1 \int f_1(x) dx + k_2 \int f_2(x) dx + \dots + k_n \int f_n(x) dx. \end{aligned}$$

To find an anti derivative of a given function, we search intuitively for a function whose derivative is the given function. The search for the requisite function for finding an anti derivative is known as integration by the method of inspection. We illustrate it through some examples.

Example 1 Write an anti derivative for each of the following functions using the method of inspection:

$$(i) \cos 2x \qquad (ii) 3x^2 + 4x^3 \qquad (iii) \frac{1}{x}, x \neq 0$$

Solution

(i) We look for a function whose derivative is $\cos 2x$. Recall that

$$\frac{d}{dx} \sin 2x = 2 \cos 2x$$

$$\text{or } \cos 2x = \frac{1}{2} \frac{d}{dx} (\sin 2x) = \frac{d}{dx} \left(\frac{1}{2} \sin 2x \right)$$

Therefore, an anti derivative of $\cos 2x$ is $\frac{1}{2} \sin 2x$.

(ii) We look for a function whose derivative is $3x^2 + 4x^3$. Note that

$$\frac{d}{dx} (x^3 + x^4) = 3x^2 + 4x^3.$$

Therefore, an anti derivative of $3x^2 + 4x^3$ is $x^3 + x^4$.

(iii) We know that

$$\frac{d}{dx}(\log x) = \frac{1}{x}, x > 0 \text{ and } \frac{d}{dx}[\log(-x)] = \frac{1}{-x}(-1) = \frac{1}{x}, x < 0$$

Combining above, we get $\frac{d}{dx}(\log|x|) = \frac{1}{x}, x \neq 0$

Therefore, $\int \frac{1}{x} dx = \log|x|$ is one of the anti derivatives of $\frac{1}{x}$.

Example 2 Find the following integrals:

$$(i) \int \frac{x^3 - 1}{x^2} dx \quad (ii) \int (x^{\frac{2}{3}} + 1) dx \quad (iii) \int (x^{\frac{3}{2}} + 2e^x - \frac{1}{x}) dx$$

Solution

(i) We have

$$\begin{aligned} \int \frac{x^3 - 1}{x^2} dx &= \int x dx - \int x^{-2} dx \quad (\text{by Property V}) \\ &= \left(\frac{x^{1+1}}{1+1} + C_1 \right) - \left(\frac{x^{-2+1}}{-2+1} + C_2 \right); C_1, C_2 \text{ are constants of integration} \\ &= \frac{x^2}{2} + C_1 - \frac{x^{-1}}{-1} - C_2 = \frac{x^2}{2} + \frac{1}{x} + C_1 - C_2 \\ &= \frac{x^2}{2} + \frac{1}{x} + C, \text{ where } C = C_1 - C_2 \text{ is another constant of integration.} \end{aligned}$$

 **Note** From now onwards, we shall write only one constant of integration in the final answer.

(ii) We have

$$\begin{aligned} \int (x^{\frac{2}{3}} + 1) dx &= \int x^{\frac{2}{3}} dx + \int dx \\ &= \frac{x^{\frac{\frac{2}{3}+1}{\frac{2}{3}+1}}}{\frac{2}{3}+1} + x + C = \frac{3}{5} x^{\frac{5}{3}} + x + C \end{aligned}$$

$$\begin{aligned}
 \text{(iii) We have } \int \left(x^{\frac{3}{2}} + 2e^x - \frac{1}{x} \right) dx &= \int x^{\frac{3}{2}} dx + \int 2e^x dx - \int \frac{1}{x} dx \\
 &= \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} + 2e^x - \log|x| + C \\
 &= \frac{2}{5} x^{\frac{5}{2}} + 2e^x - \log|x| + C
 \end{aligned}$$

Example 3 Find the following integrals:

$$\begin{aligned}
 \text{(i) } \int (\sin x + \cos x) dx & \quad \text{(ii) } \int \operatorname{cosec} x (\operatorname{cosec} x + \cot x) dx \\
 \text{(iii) } \int \frac{1 - \sin x}{\cos^2 x} dx
 \end{aligned}$$

Solution

(i) We have

$$\begin{aligned}
 \int (\sin x + \cos x) dx &= \int \sin x dx + \int \cos x dx \\
 &= -\cos x + \sin x + C
 \end{aligned}$$

(ii) We have

$$\begin{aligned}
 \int (\operatorname{cosec} x (\operatorname{cosec} x + \cot x) dx &= \int \operatorname{cosec}^2 x dx + \int \operatorname{cosec} x \cot x dx \\
 &= -\cot x - \operatorname{cosec} x + C
 \end{aligned}$$

(iii) We have

$$\begin{aligned}
 \int \frac{1 - \sin x}{\cos^2 x} dx &= \int \frac{1}{\cos^2 x} dx - \int \frac{\sin x}{\cos^2 x} dx \\
 &= \int \sec^2 x dx - \int \tan x \sec x dx \\
 &= \tan x - \sec x + C
 \end{aligned}$$

Example 4 Find the anti derivative F of f defined by $f(x) = 4x^3 - 6$, where $F(0) = 3$

Solution One anti derivative of $f(x)$ is $x^4 - 6x$ since

$$\frac{d}{dx} (x^4 - 6x) = 4x^3 - 6$$

Therefore, the anti derivative F is given by

$$F(x) = x^4 - 6x + C, \text{ where } C \text{ is constant.}$$

Given that

$F(0) = 3$, which gives,

$$3 = 0 - 6 \times 0 + C \quad \text{or} \quad C = 3$$

Hence, the required anti derivative is the unique function F defined by

$$F(x) = x^4 - 6x + 3.$$

Remarks

- (i) We see that if F is an anti derivative of f , then so is $F + C$, where C is any constant. Thus, if we know one anti derivative F of a function f , we can write down an infinite number of anti derivatives of f by adding any constant to F expressed by $F(x) + C$, $C \in \mathbf{R}$. In applications, it is often necessary to satisfy an additional condition which then determines a specific value of C giving unique anti derivative of the given function.
- (ii) Sometimes, F is not expressible in terms of elementary functions viz., polynomial, logarithmic, exponential, trigonometric functions and their inverses etc. We are therefore blocked for finding $\int f(x) dx$. For example, it is not possible to find $\int e^{-x^2} dx$ by inspection since we can not find a function whose derivative is e^{-x^2} .
- (iii) When the variable of integration is denoted by a variable other than x , the integral formulae are modified accordingly. For instance

$$\int y^4 dy = \frac{y^{4+1}}{4+1} + C = \frac{1}{5} y^5 + C$$

EXERCISE 7.1

Find an anti derivative (or integral) of the following functions by the method of inspection.

- 1. $\sin 2x$ 2. $\cos 3x$ 3. e^{2x}
- 4. $(ax + b)^2$ 5. $\sin 2x - 4 e^{3x}$

Find the following integrals in Exercises 6 to 20:

- 6. $\int (4 e^{3x} + 1) dx$ 7. $\int x^2 \left(1 - \frac{1}{x^2}\right) dx$ 8. $\int (ax^2 + bx + c) dx$
- 9. $\int (2x^2 + e^x) dx$ 10. $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 dx$ 11. $\int \frac{x^3 + 5x^2 - 4}{x^2} dx$
- 12. $\int \frac{x^3 + 3x + 4}{\sqrt{x}} dx$ 13. $\int \frac{x^3 - x^2 + x - 1}{x - 1} dx$ 14. $\int (1 - x) \sqrt{x} dx$